BTS(C) – V – 08 – 058 – O

B. Tech Degree V Semester Examination, November 2008

ME 503 MECHANICS OF MACHINERY

(2006 Scheme)

Time : 3 Hours | Maximum Marks : 100

PART A

(Answer ***ALL*** questions

(All questions carry ***EQUAL*** marks)

(8 x 5 = 40)

I. (a) Explain any two inversions of Double Slider Crank Chain.

(b) What are steering gears? Derive the fundamental equation of steering gears.

(c) Explain the concept of Forward Kinematics and Inverse Kinematics as applied to planar mechanisms.

(d) Determine the Chebychev spacing for a four bar linkage generating the function $y = 1/x$ in the range $1<=x<=2$, where three precision points are to be prescribed. Tabulate the results.

(e) State and prove law of gearing

(f) Explain the concept of interference and undercutting in gears.

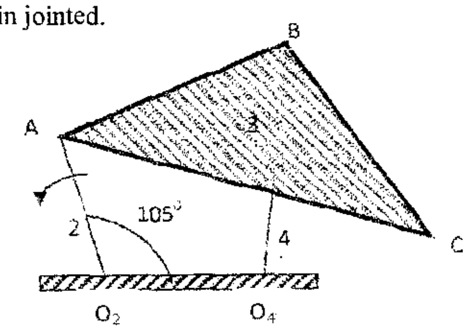
(g) Derive an expression for maximum velocity and acceleration for a follower, of a cam-follower system, executing uniform executing uniform acceleration and retardation motion.

(h) The following data relate to a screw jack : Pitch of threaded screw = 8mm, Diameter of threaded screw = 40mm, Coefficient of friction between screw and nut = 0.1, load = 20 kN. Determine the ratio of torques required to raise and lower the load.

PART B

II. (a) State and prove Kennedy's Three center's in Line Theorem. (5)

(b) Link 2 of the mechanism shown below has an angular velocity of 56 rad/s ccw. Find V_c. For the figure, AO_2 = CB=150mm, BA=BO_4 = 250mm, O_4O_2=100mm, CA = 300mm. O_4 and O_2 are pin jointed. (10)



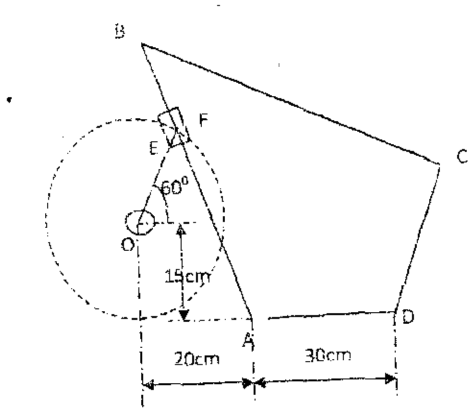
OR

III. For the mechanism shown below, the lengths of various links are OE = 15cm, AB=40cm, BC=60cm, CD=20cm, AD is the fixed link. The crank OE rotates at 70 rpm (cw). For the configuration shown, determine

(i) Velocity of die block

(ii) Coriolis component of acceleration of E with respect to F

(iii) Angular acceleration of link BC. (15)



(Turn Over

IV. Synthesise a four bar linkage to generate the function $y = \log(x)$ for $10 <= X <= 60$ using an input crank range of 120^0 and an output range of 90^0 (15)

OR

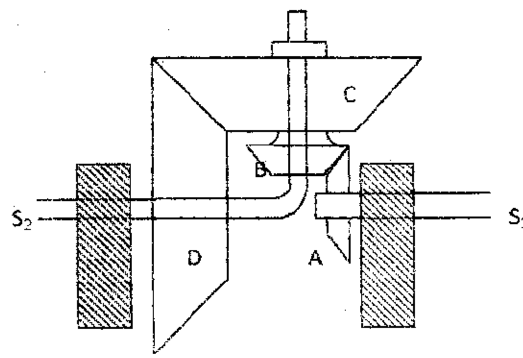
V. Derive the Jacobian relation for positioning the manipulator having three axes. (15)

VI. (a) Compare Involute profile and Cycloid profile gear tooth. (5)

(b) Two gears of module pitch 4.25 mm have 24 and 33 teeth respectively, each wheel has a standard addendum of 1 module. Determine the length of arc of contact and the maximum velocity of sliding between the mating teeth if the smaller wheel runs at 150rpm. The pressure angle is 20^0. (10)

OR

VII. In a epicyclic gear train the wheel A fixed to S_1 has 30 teeth and rotates at 500 rpm. B gears with A and is fixed rigidly to C, both being free to rotate on S_2. The wheels B,C and D have 50,70 and 90 teeth respectively. If D rotates at 80 rpm in a direction opposite to that of A, find the speed and direction of S_2 relative to S_1. (15)



VIII. It is required to set out the profile of a cam to give the following motion :

(i) Follower to move outward through an angular displacement of 20^0 during 90^0 of cam rotation

(ii) Follower to dwell for 45^0 of cam rotation

(iii) Follower to return to it's initial position during 75^0 of cam rotation

(iv) Follower to dwell through the remaining period of cam rotation.

The distance between pivot centre and follower roller centre is 70mm, Roller dia is 20 mm, minimum radius of cam is 50mm. Location of pivot point is 70 mm to the left and 60 mm above the axis of rotation of cam. The motion of follower is to take place with SHM during outstroke and with uniform acceleration and deceleration during return stroke, acceleration being 3/5 times retardation. Determine the maximum angular velocity and angular acceleration on the outstroke, if cam rotates at a uniform speed of 1200 r.p.m. (15)

OR

IX. (a) State and explain the laws of solid friction. (5)

(b) Derive an expression for displacement, velocity and acceleration of a tangent cam with roller follower, when

(i) roller is in contact with flank

(ii) roller in contact with nose. (10)

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